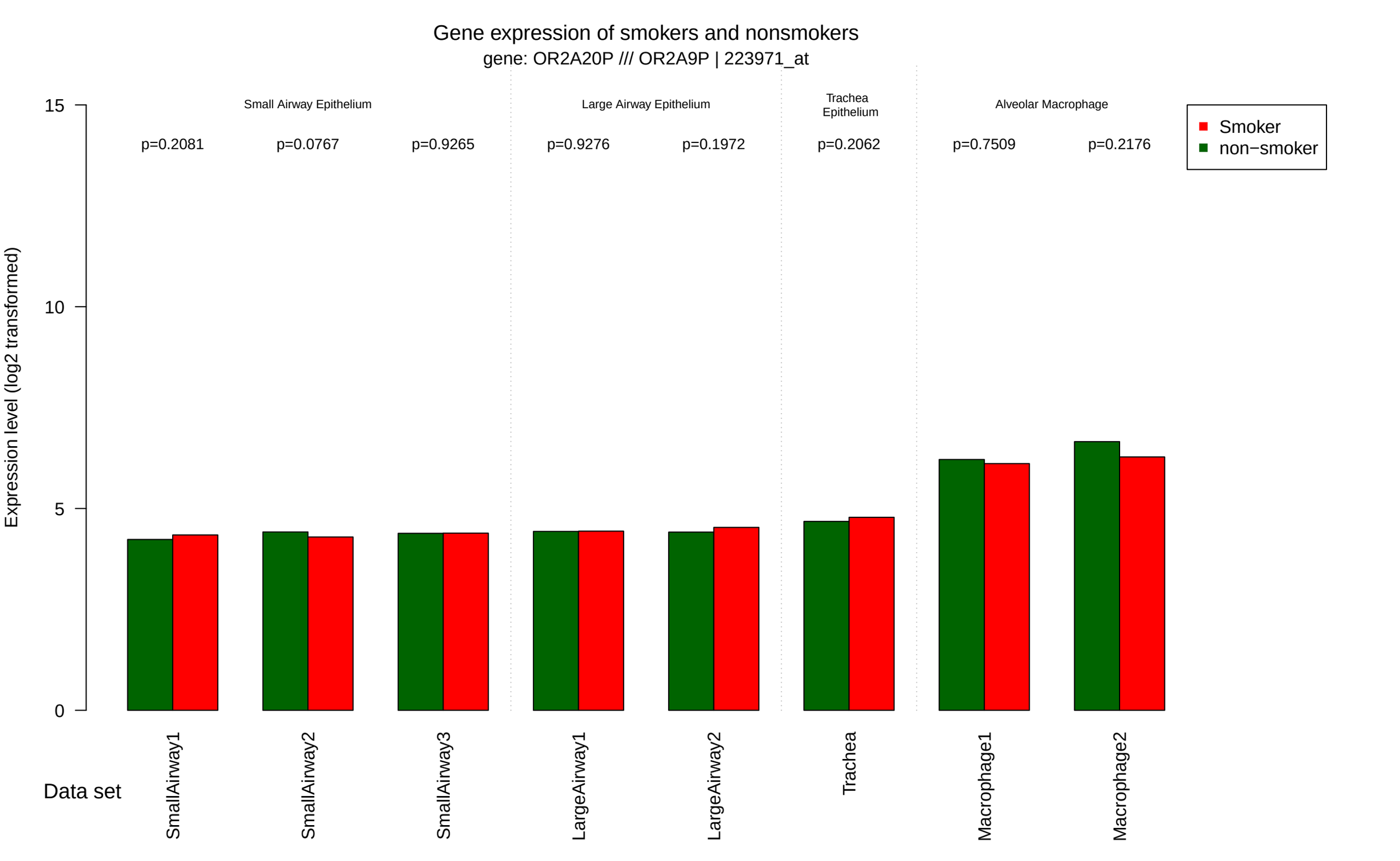
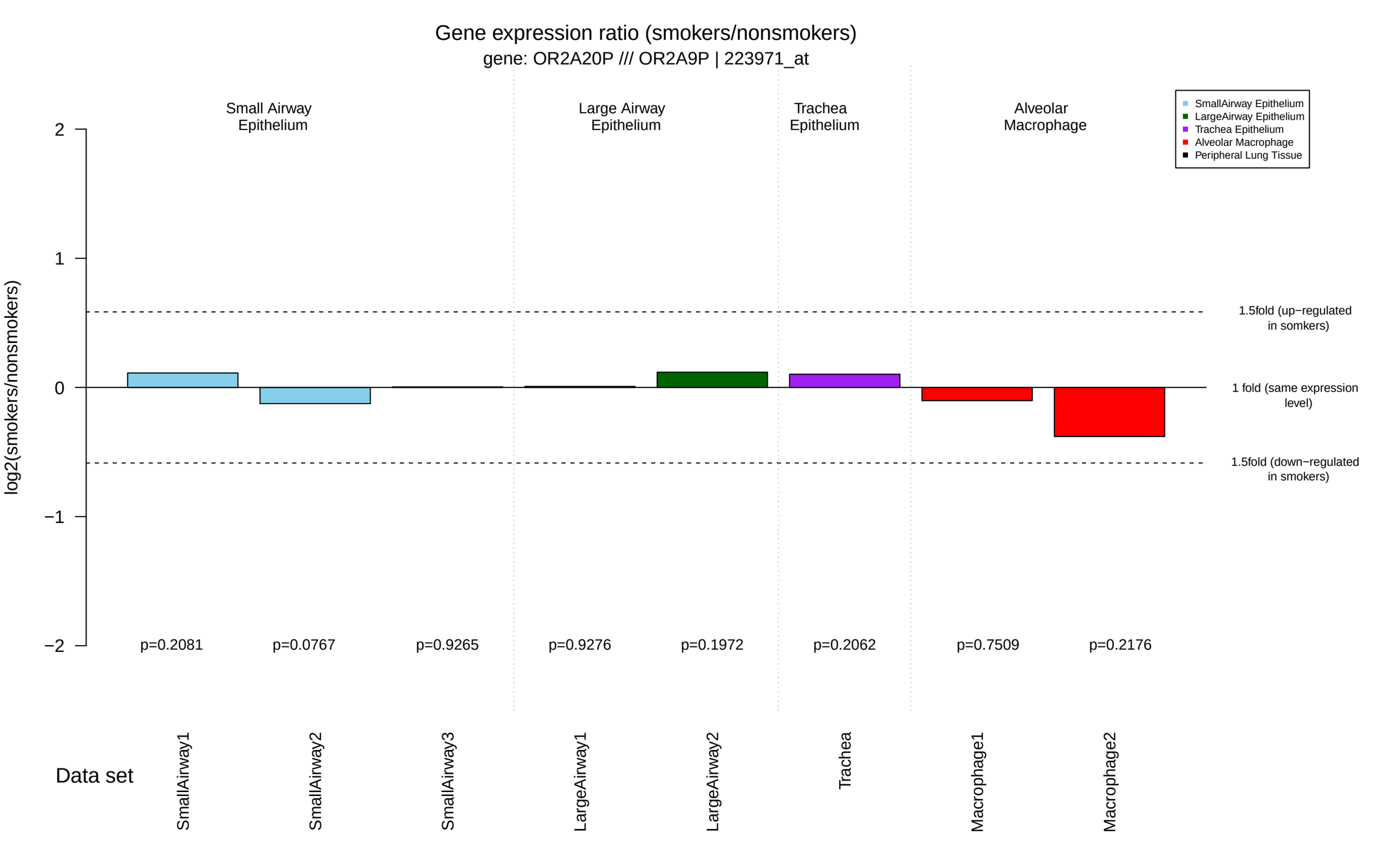
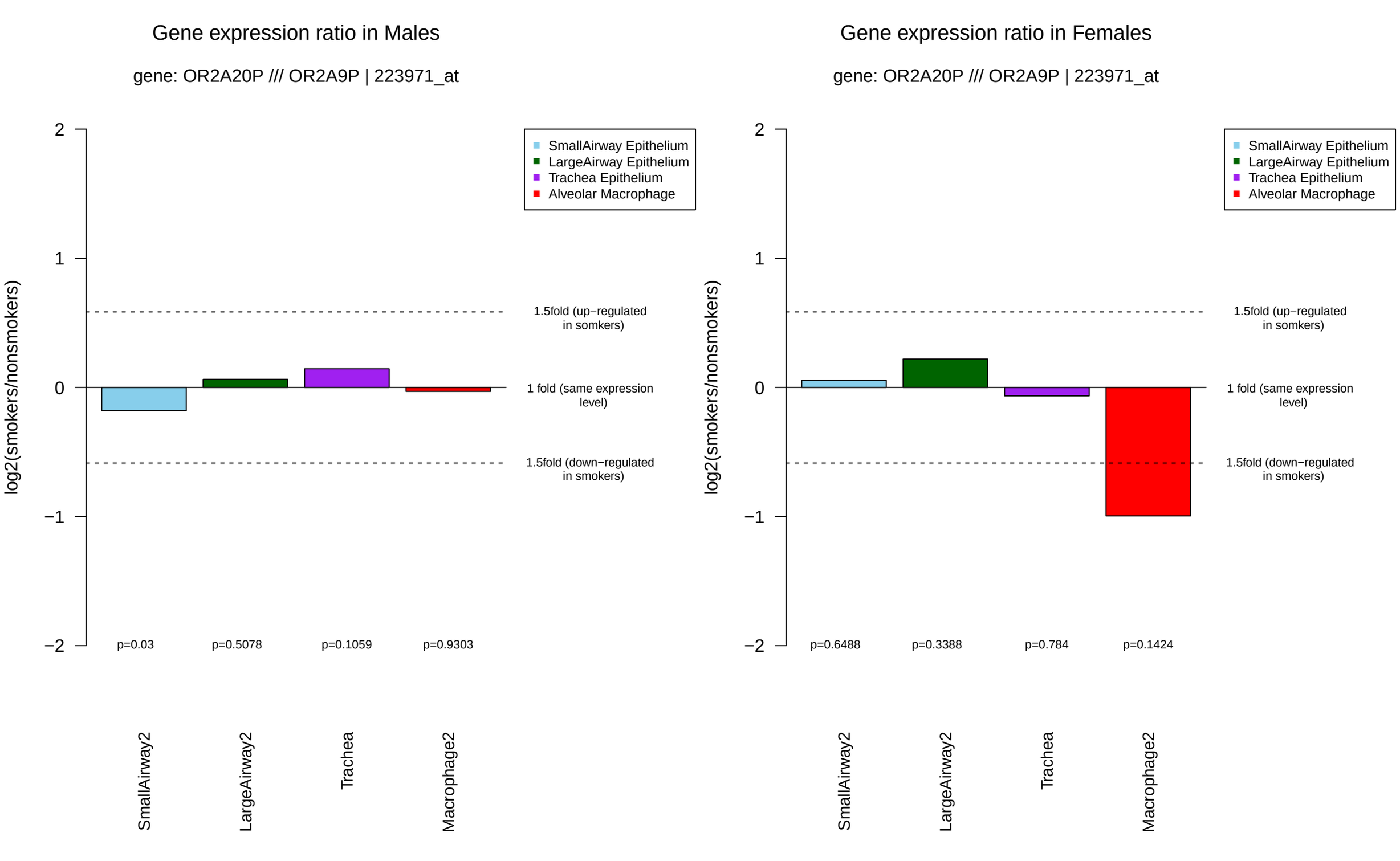
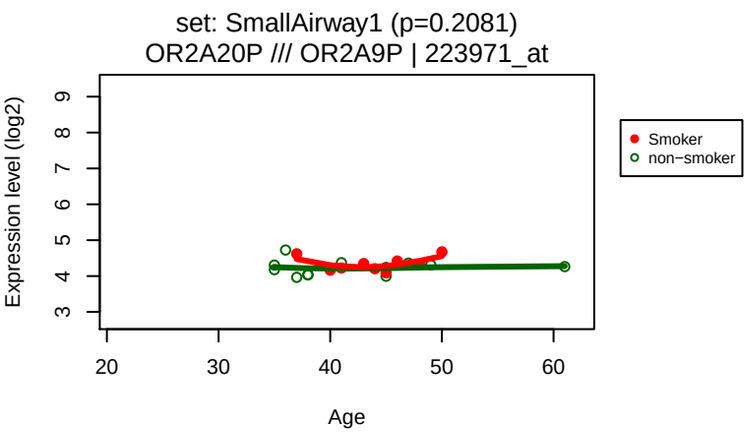
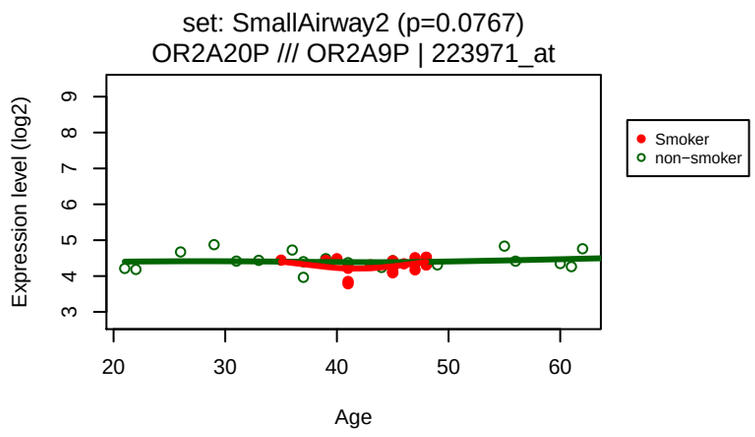
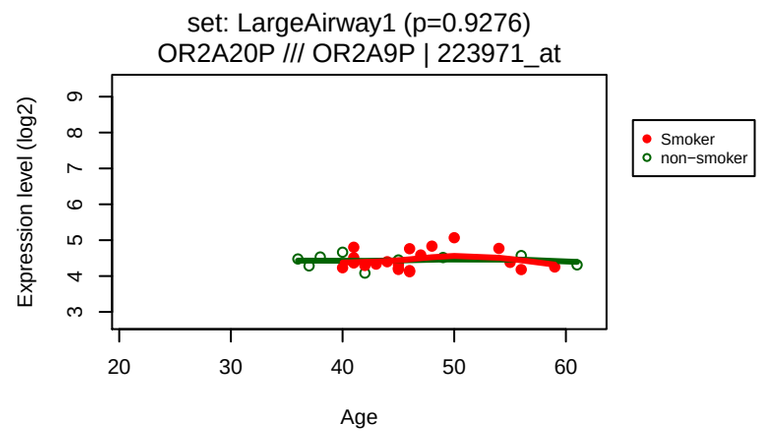

Gene expression of smokers and nonsmokers
gene: OR2A20P /// OR2A9P | 223971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2081
p=0.0767
p=0.9265
p=0.9276
p=0.1972
p=0.2062
p=0.7509
p=0.2176
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

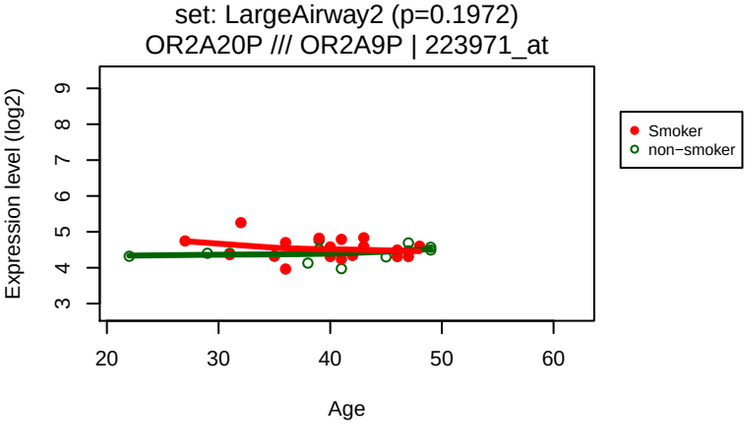

Gene expression ratio (smokers/nonsmokers)
gene: OR2A20P /// OR2A9P | 223971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2081
p=0.0767
p=0.9265
p=0.9276
p=0.1972
p=0.2062
p=0.7509
p=0.2176
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

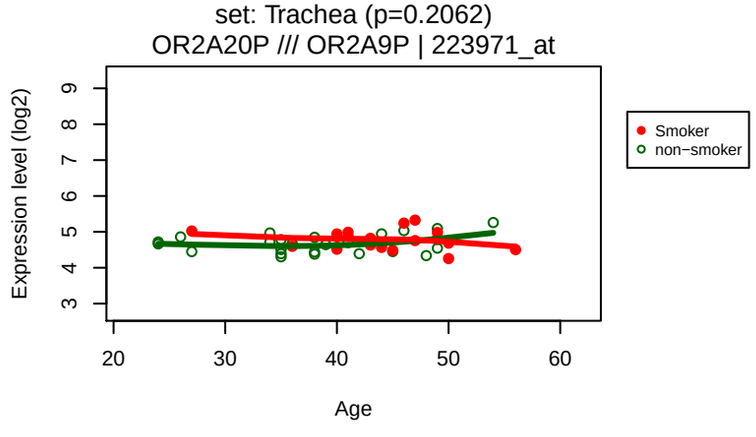

Gene expression ratio in Males
gene: OR2A20P /// OR2A9P | 223971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.03
p=0.5078
p=0.1059
p=0.9303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2A20P /// OR2A9P | 223971_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6488
p=0.3388
p=0.784
p=0.1424
SmallAirway2
LargeAirway2
Trachea
Macrophage2

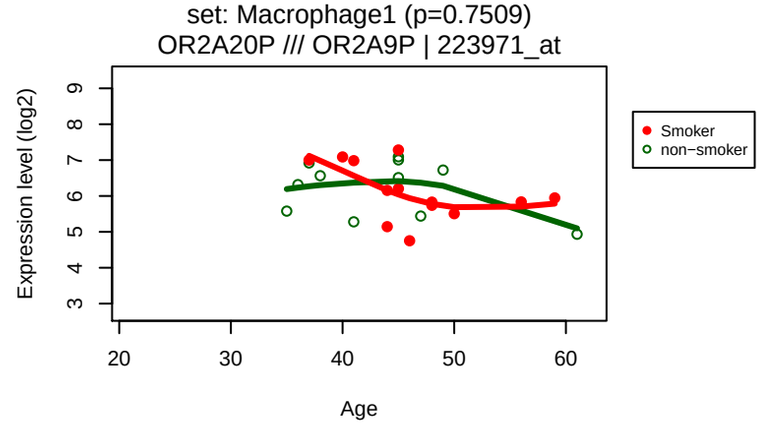

set: SmallAirway1 (p=0.2081)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker

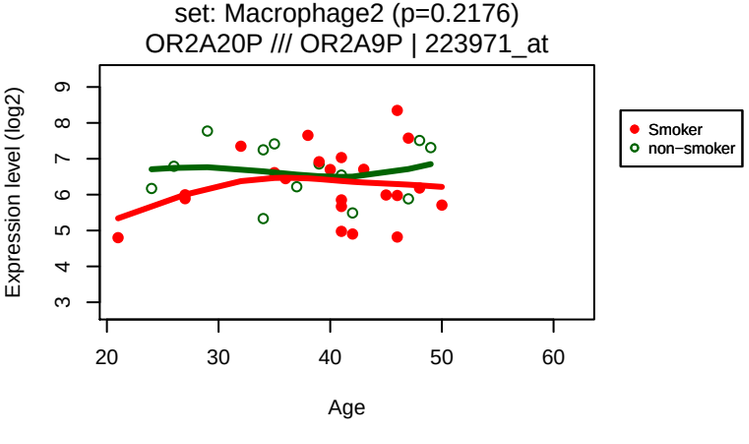

set: SmallAirway2 (p=0.0767)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker

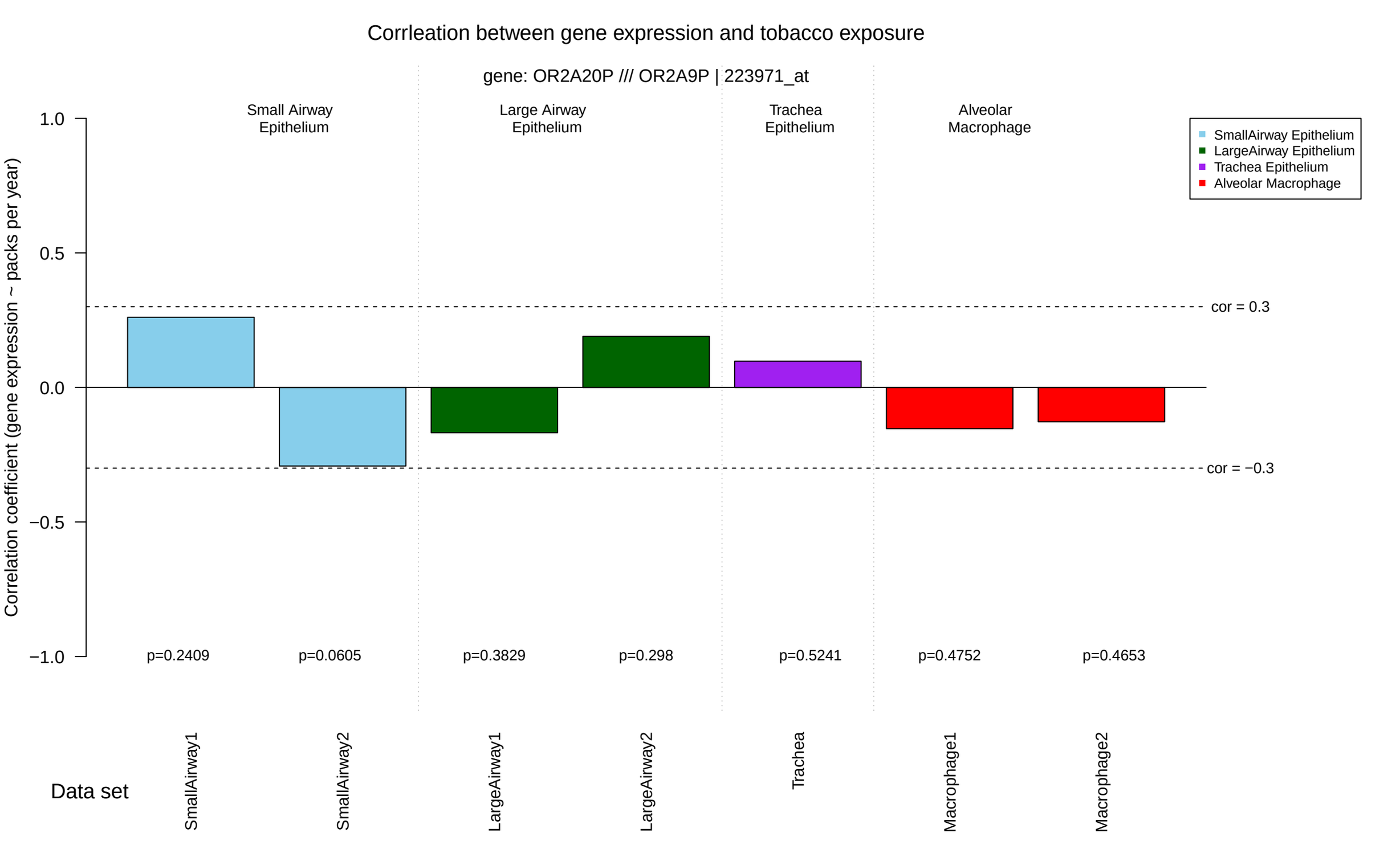

set: LargeAirway1 (p=0.9276)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1972)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2062)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7509)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2176)
OR2A20P /// OR2A9P | 223971_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2A20P /// OR2A9P | 223971_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2409
p=0.0605
p=0.3829
p=0.298
p=0.5241
p=0.4752
p=0.4653
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2